Building a Modern Financial Data Platform

You have just joined a fast-growing fintech company, ClearSpend, as a Data Engineer.

ClearSpend processes millions of credit and debit card transactions across the United States. The company partners with banks and merchants to provide analytics on customer spending behavior, merchant performance, and financial trends.

However, their current data infrastructure is fragmented.

- Transaction data is stored in raw operational exports.
- Customer and card information is maintained in separate systems.
- Merchant category data (MCC codes) exists in a messy reference file.
- There is no centralized warehouse.
- Reporting is slow, inconsistent, and unreliable.

Executives are frustrated. Finance cannot confidently report monthly revenue. The analytics team spends more time cleaning data than analysing it. Leadership wants dashboards, but the data foundation is unstable.

You have been tasked with fixing this.

Your Mission

Design and implement a complete end-to-end data pipeline that transforms raw operational data into a clean, analytics-ready data platform.

You are given four raw datasets:

1. A “transaction_data” containing card swipes and online purchases transaction-sample
2. A “users_data” containing customer information
3. A “cards_data” linking customers to payment cards
4. A “mcc_data (Merchant Category Code)” reference file

These files represent raw production exports; they are not clean, standardized, or warehouse-ready.

Your job is to: Ingest the raw data into a data platform. Clean and standardize inconsistent and messy fields. Design and build a proper data warehouse. Create data

marts that serve different finance team and the customer analytics, and the merchant partnerships teams. Ensure data quality and reliability.

The Business Context

ClearSpend leadership needs answers to questions like:

Finance Team

1. What is our total revenue by month?
2. What percentage of transactions are refunds?
3. Which states generate the most revenue?
4. Which merchant categories drive the highest spending?

Customer Analytics Team

1. What is the lifetime value of each customer?
2. How do customers behave online vs in-store?
3. How many active cards does a typical customer have?
4. Can we identify suspicious transaction patterns?

Merchant Partnerships Team

1. Which merchants generate the highest transaction volume?
2. What industries are growing the fastest?
3. Which merchants have the highest error rates?
4. How is revenue distributed geographically?

Right now, answering these questions requires manual Excel work and inconsistent logic.

You are responsible for building a system where these answers can be generated reliably and repeatedly.

What You Must Build

You are expected to design and implement:

1. An ingestion layer
2. A transformation layer
3. A dimensional data warehouse

4. Business-specific data marts

Your solution must be:

- Reproducible
- Scalable in design
- Well-modeled
- Cleanly documented

All the material for your codes and documentations must be in stored on the Github. Your report must include your design choice, diagrams, data flow, and the decision you have made throughout the project. Do not forget to add the link to the Github repo on the first page of the report.

Grading Criteria for the project:

The grades for the project will be based on 5 item (each item with 5 out of 25 for the project code and report). The 5 criteria are:

1- Architecture & Design (5 points)
2- Data Cleaning & Transformation Quality (5 points)
3- Data Warehouse Modelling (5 points)
4- Code Quality, Documentation & Reproducibility (5 points)
5- Data Marts & Business Alignment (5 points)

Here are further details:

Architecture & Design (5 points)

5 – Excellent

- Clear architecture diagram
- Logical layering (raw → transformation → curated (marts))
- Justified design decisions
- Scalable and production-oriented thinking

4 – Strong

- Good structure but minor design gaps
- Reasonable modeling choices

3 – Adequate

- Basic pipeline built but limited explanation
- Some unclear design decisions

2 – Weak

- Poor structure or missing layers
- No clear modeling logic

0–1 – Incomplete

- Major missing components

Data Cleaning & Transformation Quality (5 points)

5 – Excellent

- Handles currency formatting, negatives, nulls, and inconsistencies properly
- Strong type enforcement and validation

4 – Strong

- Most issues handled correctly
- Minor data quality gaps

3 – Adequate

- Basic cleaning performed
- Some inconsistencies remain

2 – Weak

- Superficial cleaning only

0–1 – Incomplete

- Raw data mostly unchanged

Data Warehouse Modeling (5 points)

5 – Excellent

- Proper star schema
- Clear grain definition
- Surrogate keys used
- Logical fact/dimension separation
- SCD handling where appropriate

4 – Strong

- Good schema but minor modeling weaknesses

3 – Adequate

- Functional but not well-optimized

2 – Weak

- Poor dimensional modeling

0–1 – Incomplete

- No warehouse structure

Data Marts & Business Alignment (5 points)

5 – Excellent

- Marts clearly aligned to business questions
- Aggregations well designed
- Business logic well explained

4 – Strong

- Good marts with minor clarity gaps

3 – Adequate

- Basic summaries without deeper insight

2 – Weak

- Minimal business logic

0–1 – Incomplete

- No marts built

Code Quality, Documentation & Reproducibility (5 points)

5 – Excellent

- Clean, modular, well-organized code
- README with setup instructions
- Comments explaining transformations
- Reproducible pipeline

4 – Strong

- Mostly clean and organized

3 – Adequate

- Works but messy or unclear

2 – Weak

- Hard to follow
- Poor structure

0–1 – Incomplete

- Code not runnable or missing